
RouterMAS: Value-Preserving Watermarking for Multi-Agent Routing

Google · xAI · Cornell University · Columbia University

Research extended abstract

Abstract

Multi-agent systems derive capability from how they delegate work, yet embedding an ownership signal into delegation can alter every decision that follows. Existing behavioral and expert-routing watermarks provide signature carriers without characterizing the routing freedom left once a system must preserve its continuation utility and cost. We propose RouterMAS, a value-preserving watermarking framework for multi-agent routing. Its key insight is that fixing task values need not fix the routing distribution: Bellman recursion converts prefix-wise preservation into local conservation equations, exposing an admissible space of probability redistributions. RouterMAS constructs this space from continuation values, encodes a secret key through a minimum-relative-entropy routing law, and verifies the signature with sequential likelihood-ratio evidence induced by the same law. An error budget connects value estimation and deployed-policy approximation to accumulated return drift, and a single projected component governs both the keyed update and its expected verification information. In a controlled multi-agent routing system with exact dynamics, the marked router preserves both protected values to 10^{-14} while changing the routing law by 0.093 nats per decision; a law that matches only first-step rewards leaves downstream drift of 0.016 in utility and 0.035 in cost; row-space keys and full-rank histories are inert; and the running-maximum test detects 99.3% of marked trace groups within 128 decisions at a false-positive rate of 0.75%. We specify the evaluation protocol for reasoning, code-generation and environment-interaction multi-agent systems.

1 Introduction

Language-model agents increasingly solve tasks by proposing, checking, revising and executing one another's work. Role-based collaboration structures these interactions [Hong et al., 2024; Chen et al., 2024], while learned routers select collaboration modes, roles and constituent models [Yue et al., 2025]. Two systems built from the same models can therefore embody different orchestration policies. The routing policy is a system-level asset, and communication-topology inference exposes a corresponding provenance problem [Wu et al., 2026].

A routing watermark must change something a verifier can observe, but delegation is also a functional decision. State-conditioned orchestration treats agent selection as a sequential decision process [Dang et al., 2025]: a delegation changes the messages and history on which later choices, retries and costs depend. We study a deployment contract that preserves expected task utility and operating cost *after every admissible execution prefix*. The contract prevents gains on easy prefixes from compensating for losses on hard ones, and it makes a fixed secret path an inadequate starting point: such a path is identifiable precisely because it sends work where the original system would not.

Text watermarks bias generated tokens [Kirchenbauer et al., 2023]; behavioral watermarks encode agent decisions [Huang et al., 2026]; trajectory watermarks target downstream learning [Meng et al., 2026]; and mixture-of-experts watermarks use triggered expert paths [Gao et al., 2026]. Equivalent-action and history-conditioned watermarks also exploit execution structure [Wang et al., 2026; An et al., 2026]. Transferred to multi-agent routing, these *carrier-first* designs share an implicit assumption: that a secret preference can be fixed before its continuation-value admissibility is established. The missing object is the admissible carrier itself. A delegation that looks

harmless now need not be harmless downstream, and a history whose routing probabilities are uniquely determined by the protected values leaves no room for a value-preserving intervention.

The key is therefore not to choose a watermark route, but to identify which routing variations remain undetermined by the values the system must preserve. We ask: *which multi-agent routing variations preserve required continuation values, and how can they carry a verifiable signature with controlled intervention and estimation cost?* RouterMAS answers in two stages. Embedding constructs the value-preserving space and encodes the key within it; verification accumulates evidence from the resulting delegations. The router is returned unchanged where no admissible signal remains, and an explicit error budget governs the estimated and deployed versions.

Our contributions are threefold. (1) We formulate system-level multi-agent router watermarking as probability redistribution under prefix-wise continuation-value preservation and derive the exact local feasible family, including its zero-freedom boundary. (2) We derive the keyed least-change encoder and show that the same probability-weighted projection sets both its local score gain and its expected verification information, with an anytime-valid sequential test. (3) We verify the mechanism in a controlled routing system with exact dynamics and specify the evaluation protocol for LLM multi-agent systems.

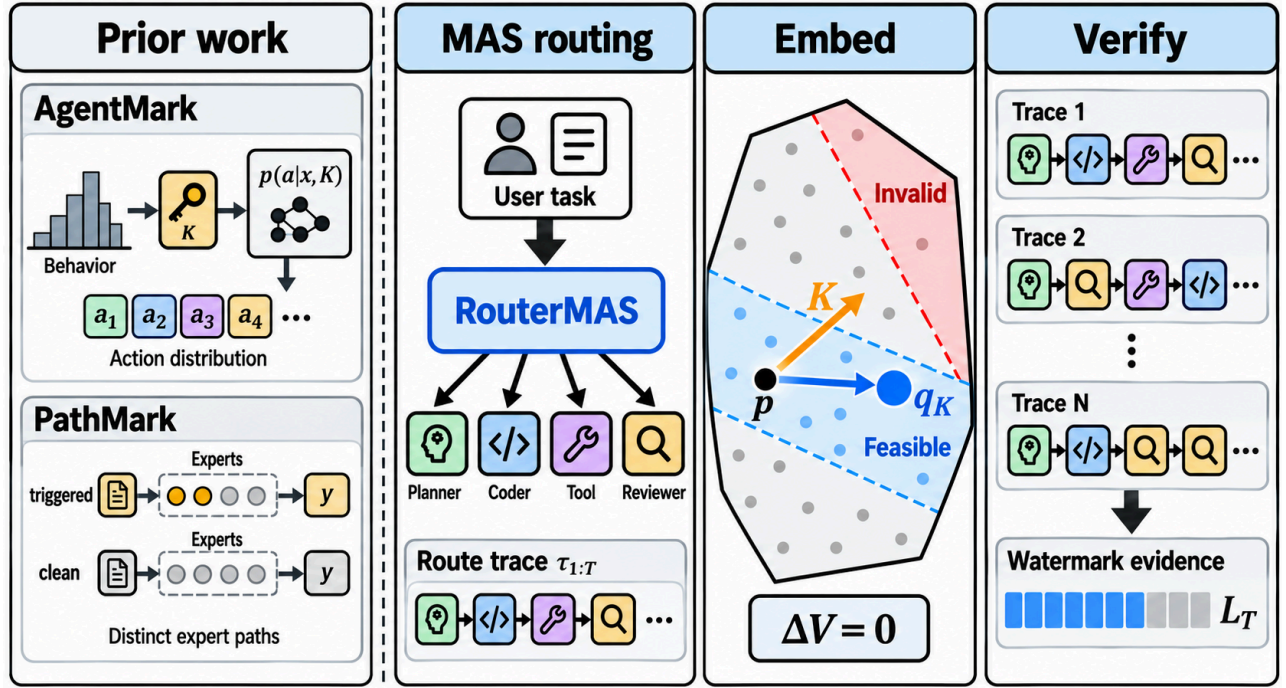


Figure 1: RouterMAS and prior carriers. Behavioral watermarks mark an agent’s action distribution and path watermarks mark triggered expert routes. RouterMAS marks the system-level routing policy: a private key shifts p to q_K within the continuation-value-preserving region ($\Delta V = 0$), and delegation traces accumulate likelihood-ratio evidence L_T .

2 Problem Setting

We consider a finite-horizon multi-agent system with frozen constituent agents, tools and conditional executor. At history h the legal delegation set is $\mathcal{A}_h$ and the reference routing law $p_h = p(\cdot | h)$ is positive on it. A delegation specifies an agent together with its role or communication destination; the executor produces a response and the next history, and the horizon H includes an absorbing termination state. For each of m protected returns j , $V_j^p(h)$ is the remaining expected return under p and $Q_j^p(h, a)$ the return after delegating to a and then following p . Task success and remaining agent-call cost supply the two principal returns. The contract is $V_j^q(h) = V_j^p(h)$ for every admissible prefix and every protected return.

The owner controls routing and retains the reference policy. A precommitted secret key K determines a bounded delegation score $s_{K,h}$, and the marked law $q_{K,h}$ is deployed with the router. A verifier reads delegation histories and reconstructs the reference and marked conditional laws from frozen snapshots. The level α is the reference-null false-positive budget and L_T the accumulated log evidence over T auditable decisions.

3 Method

3.1 Continuation conservation defines the carrier

Algorithm 1 audits each legal delegation with complete counterfactual continuations under the unchanged reference router. Following a selected delegation to termination is essential, because the preserved quantity includes the future decisions its response enables. Constrained policy optimization separates protected returns from the search objective [Achiam et al., 2017]; requiring preservation at every prefix gives an exact characterization.

Proposition 1 (Prefix-wise conservation). *With a shared finite-horizon executor and terminal return convention, $V_j^q(h) = V_j^p(h)$ at every admissible history if and only if $(q_h - p_h)^\top Q_j^p(h) = 0$ at every history, for each protected return j .*

Backward induction proves sufficiency: at depth $H - 1$ the shared terminal convention gives $Q_j^q = Q_j^p$, local conservation gives equal history values, and the equality propagates upward through the shared executor. Equal successor values make the two Bellman recursions differ only by the local term, which proves necessity. Stacking normalization and the protected values into $B_h = [\mathbf{1}^\top; Q_1^p(h)^\top; \dots; Q_m^p(h)^\top]$ gives the affine family $\mathcal{F}_h = \{q \geq 0 : B_h(q - p_h) = 0\}$. Because p_h is strictly positive, every $d \in \ker B_h$ admits small moves $p_h \pm td$, so

$$\dim \mathcal{F}_h = |\mathcal{A}_h| - \text{rank } B_h. \quad (1)$$

Full column rank leaves only p_h . If a key score lies in the row space, $s = B_h^\top v$, then $s^\top(q - p_h) = 0$ for every feasible q and the keyed objective reduces to $-\eta \text{KL}(q||p_h)$, whose maximizer is p_h . These two cases define the exact no-marking branch.

Algorithm 1 Value-preserving watermark embedding

Require: history h , reference router, frozen executor, protected returns, key K , audit budget M , weight $\eta > 0$, budgets τ_j

1. $(\mathcal{A}, p) \leftarrow \text{LegalRouting}(h)$
 2. $(\widehat{Q}, \epsilon) \leftarrow \text{ContinuationAudit}(h, \mathcal{A}, M)$
 3. $B \leftarrow [\mathbf{1}^\top; \widehat{Q}_1^\top; \dots; \widehat{Q}_m^\top]$
 4. $\mathcal{F} \leftarrow \{q \geq 0 : B(q - p) = 0\}$
 5. $D \leftarrow \text{diag}(p)$
 6. $P \leftarrow \text{OrthogonalProjector}(\ker(BD^{1/2}))$
 7. $s_K \leftarrow \text{KeyScore}(K, h, \mathcal{A})$
 8. $g \leftarrow D^{1/2}PD^{1/2}s_K$
 9. **if** $g = 0$ or the audit error is unbounded **return** p
 10. $q_K^* \leftarrow \arg \max_{q \in \mathcal{F}} \{s_K^\top(q - p) - \eta \text{KL}(q||p)\}$
 11. $\delta \leftarrow q_K^* - p$; $b_j \leftarrow |\delta^\top \widehat{Q}_j|$
 12. $\rho \leftarrow \min\{1, \min_j \tau_j / (b_j + \epsilon_j \|\delta\|_1)\}$
 13. **return** $q_K \leftarrow p + \rho\delta$
-

Algorithm 2 Sequential routing-signature verification

Require: auditable trace $\{(h_t, a_t)\}_{t < T}$, frozen snapshots, precommitted K , level α

1. $L_0 \leftarrow 0$
 2. **for** $t = 0, \dots, T - 1$ **do**
 3. $(p_{h_t}, q_{K,h_t}) \leftarrow \text{ReconstructLaws}(h_t, K)$
 4. $L_{t+1} \leftarrow L_t + \log\{q_{K,h_t}(a_t)/p_{h_t}(a_t)\}$
 5. **if** $L_{t+1} \geq \log(1/\alpha)$ **return** detected
 6. **return** not detected at budget T
-

3.2 Least-change encoding inside the feasible family

For an attainable increase in expected key score, encoding selects the feasible law with minimum relative entropy to the reference, an information projection [Csiszár, 1975]. With a common initial-task law and shared executor, trajectory KL equals the marked-policy expectation of the summed conditional routing KLs, which gives the

penalty an additive sequential meaning. The carrier itself is divergence-independent; KL supplies this accounting and an exponential-family solution. Stationarity of the Lagrangian gives

$$q_K(a) = \frac{p(a)}{Z(\lambda)} \exp\left(\frac{s_K(a) - \lambda^\top \widehat{Q}_a}{\eta}\right), \quad \Phi(\lambda) = \eta \log \sum_a p(a) e^{(s_K(a) - \lambda^\top \widehat{Q}_a)/\eta} + \lambda^\top \widehat{Q}p, \quad (2)$$

where Φ is a convex dual with gradient $\widehat{Q}p - \widehat{Q}q(\lambda)$ and Hessian $\eta^{-1} \text{Cov}_{q(\lambda)}(\widehat{Q}_a)$; relative-entropy policy updates share this structure [Schulman et al., 2015]. Expanding $\text{KL}(p + \delta \| p) = \frac{1}{2} \delta^\top D^{-1} \delta + O(\|\delta\|^3)$ equips probability changes with the Fisher metric of natural policy gradients [Kakade, 2001], and the constrained projection yields the local direction $\eta^{-1}g$.

3.3 Error budget

For an implemented router $\tilde{q}$, let J_j be the initial-task expected return, $b_{j,h} = |(\tilde{q}_h - p_h)^\top \widehat{Q}_j(h)|$ the implemented residual, and $\epsilon_{j,h}$ a maximum-norm bound on the continuation error. The finite-horizon performance-difference identity gives

$$|J_j(\tilde{q}) - J_j(p)| \leq \mathbb{E}_{\tilde{q}} \sum_{t < H} \left[b_{j,h_t} + \epsilon_{j,h_t} \|\tilde{q}_{h_t} - p_{h_t}\|_1 \right], \quad (3)$$

with the expectation over the implemented policy’s own visits. Audit uncertainty, numerical residuals and controller compilation enter one return budget. The safe-scaling factor ρ in Algorithm 1 keeps each local charge within τ_j ; since $p + \rho\delta$ stays on the segment to a feasible proposal, scaling preserves nonnegativity and estimated conservation.

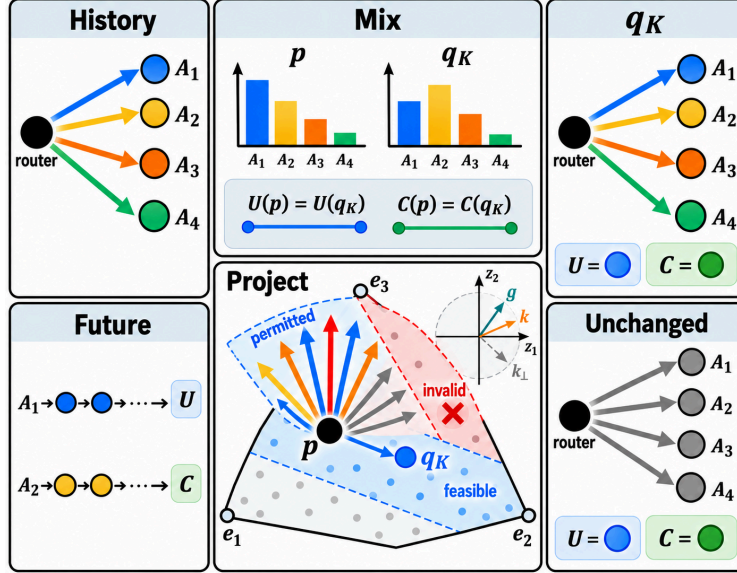


Figure 2: Value-preserving routing at one history. p and q_K mix the delegations differently with equal continuation utility and cost; value-changing directions are excluded, and histories without a usable direction keep the reference router.

3.4 One projection sets the update and its evidence

Corollary 1 (Local embedding and evidence). For positive reference support and fixed constraints and key score, $s_K^\top g = g^\top D^{-1}g = \|PD^{1/2}s_K\|^2$. As $\eta \rightarrow \infty$ the exact unscaled optimizer satisfies $s_K^\top(q_K^* - p) = \eta^{-1}s_K^\top g + O(\eta^{-2})$ and $\mathbb{E}_{a \sim q_K^*} \log \frac{q_K^*(a)}{p(a)} = \frac{1}{2}\eta^{-2}s_K^\top g + O(\eta^{-3})$.

The same projected component therefore selects the keyed update and determines its leading-order verification information; zero projection makes both inert. For the returned law $p + \rho(q_K^* - p)$ the leading score term scales by ρ and the leading evidence term by ρ^2 .

3.5 Verification from the marked routing law

Algorithm 2 reconstructs both conditional laws for each observed delegation and accumulates their likelihood ratio, as in likelihood-ratio watermark detection [Hu et al., 2024]. Under the conditional-action null p , $\mathbb{E}_p[q_{K,h_t}(a_t)/p_{h_t}(a_t) \mid h_t] = 1$, so $\exp(L_t)$ is a nonnegative martingale and Ville's inequality bounds the probability that its running maximum ever reaches $1/\alpha$ by α [Howard et al., 2020]. The verifier may stop at the first crossing. Under the marked law the conditional expected increment is $\text{KL}(q_{K,h} \parallel p_h)$, the verification side of Corollary 1. Key selection and threshold calibration precede inspection of test traces; testing several keys requires family-wise calibration.

4 Controlled Evidence with Exact Dynamics

A finite-state multi-agent routing system isolates the mechanism from estimation error: four delegation steps, five states per step, six legal delegations at every history and two additive protected returns (task utility and agent-call cost). One seed (20260923) draws transition kernels, rewards, reference probabilities and the key; backward dynamic programming gives exact continuation values; the encoder uses $\eta = 1.5$. A separate three-action instance yields full-rank constraints. Every number below was produced by re-executing these checks.

Table 1: Conservation and inert controls (maxima over 20 histories; mean KL per decision).

Routing law	$\max \ \mathcal{B}\delta\ _\infty$	$\max \Delta V_U $	$\max \Delta V_C $	Mean KL
Unmarked	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	0
Unprojected key bias	0.2037	0.2432	0.2304	0.1666
Immediate-value constrained	0.0369	0.0156	0.0352	0.0898
Random feasible direction	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	0.0065
Row-space key	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	0
Full-rank (3 actions)	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	0
RouterMAS	$< 10^{-14}$	$< 10^{-14}$	$< 10^{-14}$	0.0929

ΔV is the change in history value under the marked law. Values below 10^{-14} are floating-point rounding.

Continuation constraints preserve both protected values to rounding precision while changing the routing law (Table 1). Constraining only first-step rewards satisfies a weaker local condition and leaves downstream drift; at the last step, where nothing follows, the two coincide. A random feasible direction preserves values but carries little key information, and the two inert controls return the reference law exactly.

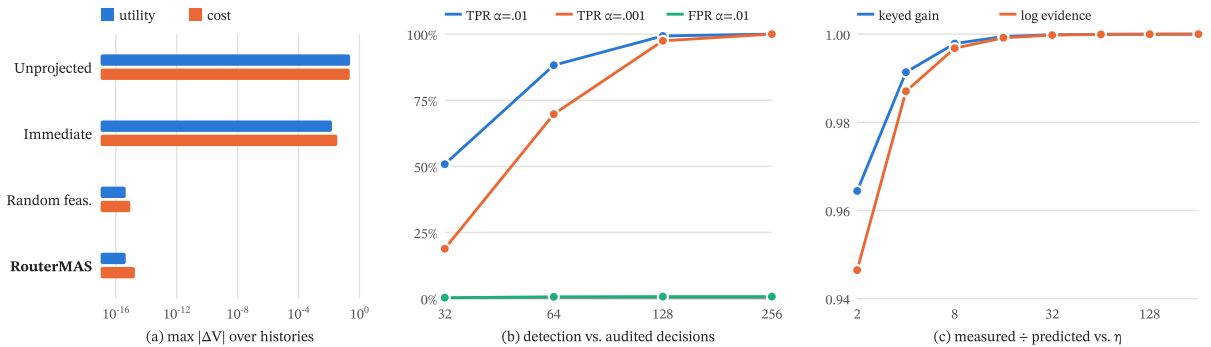


Figure 3: Executed checks. (a) Worst prefix value drift by routing law, log scale. (b) Running-maximum detection over 10,000 independent trace groups per law. (c) Measured ÷ predicted keyed gain and expected log evidence across the 20 histories as η grows (Corollary 1).

Table 2: Sequential verification (%).

T	FPR _{.01}	TPR _{.01}	FPR _{.001}	TPR _{.001}
32	0.36	50.81	0.02	18.88
64	0.67	88.24	0.03	69.73
128	0.75	99.30	0.04	97.46
256	0.75	100.00	0.05	99.99

Table 3: Embedding–evidence relation.

η	gain coef.	evidence coef.	max residual
2	0.96442	0.94647	1.2×10^{-15}
8	0.99786	0.99680	5.0×10^{-16}
32	0.99985	0.99977	1.1×10^{-15}
128	0.99998	0.99998	1.4×10^{-15}
256	0.99999	0.99999	9.2×10^{-16}

Longer traces increase power while empirical false-positive rates stay below their nominal levels (Table 2). With all 20 histories, the key and p , B frozen, the normalized coefficients of Corollary 1 approach one as η grows (Table 3); the shared quadratic form holds to 2.2×10^{-16} . Halving the final update ($\rho = 0.5$, $\eta = 128$) gives a score ratio of 0.5000 and an evidence ratio of 0.2500, and an independent categorical replay with 10^6 draws per history agrees with the exact KL within 1.34 standard errors. An independent prototype with estimated values (128 complete reference rollouts per delegation, 15,360 counterfactual continuations, local budget $\tau = 0.01$) realizes initial drift of 2.0×10^{-4} and 2.5×10^{-4} against the Equation (3) budget of 0.040 and 0.026. These are controlled numerical results; they verify the mechanism rather than performance on LLM routers.

5 Evaluation Protocol for LLM Multi-Agent Systems

The controlled system establishes the mechanism with exact values. The empirical question for deployed routers is whether they expose enough structured residual freedom for calibrated detection at low utility and cost drift, and how much of the marked law survives modification. We fix the following protocol before measurement.

Tasks and systems. GSM8K for mathematical reasoning [Cobbe et al., 2021], HumanEval and MBPP for code generation [Chen et al., 2021; Austin et al., 2021] and ALFWorld for environment interaction [Shridhar et al., 2021]. The primary system uses six role-specialized Qwen2.5-7B-Instruct delegates [Qwen Team, 2024] (planner, solver, critic, verifier, tool specialist, synthesizer) under an eight-delegation horizon; Llama-3.1-8B-Instruct [Grattafiori et al., 2024] supplies a second model family. Centralized, layered and graph-routed configurations share the frozen executor within each comparison. Controller fitting, value fitting, threshold calibration and testing are disjoint by root task, and every counterfactual branch stays with its root task.

Comparisons. Unmarked routing; delegation-level red/green bias [Kirchenbauer et al., 2023]; a behavioral-watermark adaptation at the delegation level [Huang et al., 2026]; a forced inter-agent path inspired by expert-path marking [Gao et al., 2026]; and the mechanism controls of Section 4 (immediate-value projection, a global-average penalty, random feasible redistribution). All methods share the action alphabet, executor, observation interface and decision budget.

Metrics. Task success or pass@1; all-episode modeled cost normalized to the unmarked system; $D_{U,95}$, the 95th-percentile absolute continuation-utility drift over held-out audit prefixes; TPR at an independently validated FPR for $T \in \{32, 64, 128, 256\}$ auditable decisions, primary $\alpha = 0.01$, three execution seeds and ten precommitted keys. Continuation audits use 32 rollouts per audited delegation in a pilot and compare 8, 32, 128 and 512 in the expanded audit.

Hypotheses and falsification. *H1 (freedom exists):* held-out prefixes have nonzero feasible dimension and nonzero projected key energy; if the protected returns determine the routing law, RouterMAS abstains and the method does not apply to that router. *H2 (value-preserving detection):* at matched decision budgets RouterMAS reaches calibrated detection with success within one percentage point and modeled cost within three percent of the unmarked system, and lower $D_{U,95}$ than carriers without continuation constraints. *H3 (geometry):* on frozen prefixes with estimated values, the normalized gain and evidence coefficients of Corollary 1 approach one as η

grows. Persistence is measured under copying, reversible relabeling, deterministic decoding, fine-tuning, imitation and value-preserving erasure, each reported with its access model.

6 Related Work

Multi-agent orchestration. Role-based frameworks organize collaboration among language agents [Hong et al., 2024; Chen et al., 2024]; learned routers choose collaboration modes, roles and models [Yue et al., 2025]; and evolving orchestration models delegation as a sequential decision process [Dang et al., 2025]. Topology inference shows that orchestration structure can leak [Wu et al., 2026]. RouterMAS treats the resulting routing policy as the object of provenance.

Watermarking decisions and routes. Token-level watermarks bias generation toward keyed sets and test with likelihood statistics [Kirchenbauer et al., 2023; Hu et al., 2024]. Behavioral watermarks embed signatures in agent action distributions [Huang et al., 2026], related schemes exploit equivalent actions and history-conditioned behavior [Wang et al., 2026; An et al., 2026], trajectory watermarks target models trained on agent traces [Meng et al., 2026], and path watermarks use triggered expert routes in mixture-of-experts models [Gao et al., 2026]. These works establish that decisions and routes can carry provenance. RouterMAS supplies the admissible carrier for a multi-agent router: it derives which routing changes preserve continuation values before any key is applied, and abstains where none exist.

Constrained and information-geometric policy updates. Constrained policy optimization [Achiam et al., 2017], trust-region updates [Schulman et al., 2015] and natural gradients [Kakade, 2001] inform the error budget and the local geometry; the encoder is an information projection [Csiszár, 1975], and the verifier is an anytime-valid likelihood-ratio test [Howard et al., 2020].

7 Conclusion

RouterMAS places a multi-agent watermark in the routing freedom left by required continuation values. Bellman conservation defines the carrier, relative entropy determines the keyed redistribution, and the resulting law supplies sequential evidence through the same projection. Controlled experiments with exact dynamics confirm conservation, the failure of immediate-value matching, the inert boundary and calibrated detection. This perspective makes the task-preserving degrees of freedom of an orchestration policy an explicit object of provenance design.

References

- Achiam, J., Held, D., Tamar, A., and Abbeel, P. Constrained policy optimization. *ICML*, 2017.
- An, H., Park, S., Kim, D., and Han, Y.-S. Sequential behavioral watermarking for LLM agents. arXiv:2605.11036, 2026.
- Austin, J. et al. Program synthesis with large language models. arXiv:2108.07732, 2021.
- Chen, M. et al. Evaluating large language models trained on code. arXiv:2107.03374, 2021.
- Chen, W., Su, Y., Zuo, J., Yang, C., et al. AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors. arXiv:2308.10848, 2024.
- Cobbe, K. et al. Training verifiers to solve math word problems. arXiv:2110.14168, 2021.
- Csiszár, I. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975.
- Dang, Y., Qian, C., Luo, X., et al. Multi-agent collaboration via evolving orchestration. *NeurIPS*, 2025.
- Gao, Y., Wang, Q., Yuan, Y., et al. PathMark: Protecting intellectual property of mixture-of-expert LLMs via path watermarks. *ACM CCS*, 2026.
- Grattafiori, A. et al. The Llama 3 herd of models. arXiv:2407.21783, 2024.
- Hong, S., Zhuge, M., Chen, J., et al. MetaGPT: Meta programming for a multi-agent collaborative framework. arXiv:2308.00352, 2024.
- Howard, S. R., Ramdas, A., McAuliffe, J., and Sekhon, J. Time-uniform Chernoff bounds via nonnegative supermartingales. *Probability Surveys*, 17:257–317, 2020.

Hu, Z., Chen, L., Wu, X., Wu, Y., Zhang, H., and Huang, H. Unbiased watermark for large language models. arXiv:2310.10669, 2024.

Huang, K., Tan, J., Wei, Y., et al. AgentMark: Utility-preserving behavioral watermarking for agents. *ACL*, 2026.

Kakade, S. M. A natural policy gradient. *NeurIPS*, 2001.

Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., and Goldstein, T. A watermark for large language models. *ICML*, 2023.

Meng, W., Gong, C., Zhuo, T. Y., et al. Watermarking LLM agent trajectories. *ICML*, 2026.

Qwen Team. Qwen2.5 technical report. arXiv:2412.15115, 2024.

Schulman, J., Levine, S., Moritz, P., Jordan, M. I., and Abbeel, P. Trust region policy optimization. *ICML*, 2015.

Shridhar, M., Yuan, X., Côté, M.-A., Bisk, Y., Trischler, A., and Hausknecht, M. ALFWorld: Aligning text and embodied environments for interactive learning. arXiv:2010.03768, 2021.

Wang, L., Li, Z., Xie, Y., et al. On protecting agentic systems' intellectual property via watermarking. arXiv:2602.08401, 2026.

Wu, Y., Lin, X., Zhang, H., et al. CIA: Inferring the communication topology from LLM-based multi-agent systems. *ACL*, 2026.

Yue, Y., Zhang, G., Liu, B., et al. MasRouter: Learning to route LLMs for multi-agent systems. *ACL*, 2025.